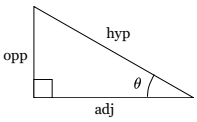
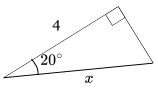
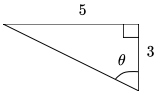
1.39 Trigonometry: Introduction

Chapter 9 · Original slide 4 · RECAP: Right-Angled Trigonometry





Find x.



Find θ.

You are probably familiar with the formula: $\sin(\theta) = \frac{opp}{hyp}$

But what is the *conceptual* definition of $\sin$?

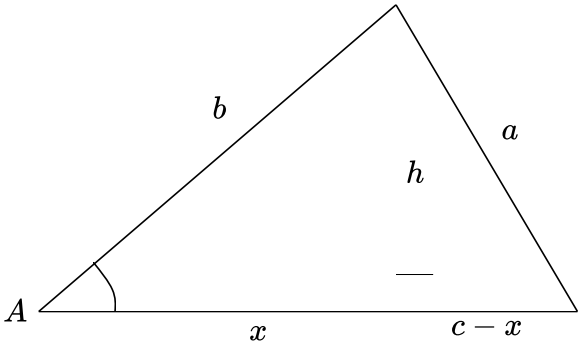
sin is a function which inputs an angle and gives the ratio between the opposite and hypotenuse.

Remember that a ratio just means the 'relative size' between quantities (in this case lengths). For this reason, sin/cos/tan are known as "trigonometric ratios".

reflections: You may have been taught "use tan^{-1} whenever you're finding an angle", and therefore write the second line directly. It is preferable to always write the first line, then see the problem as a 'changing the subject' one. We need to remove the tan on front of the θ, so apply tan^{-1} to each side of the equation to 'cancel out' the tan on the LHS.

Tip: You can swap the thing you're dividing by and the result. e.g. $\frac{8}{2} = 4 \rightarrow \frac{8}{4} = 2$.

Chapter 9 · Original slide 5 · Proof of Cosine Rule



We want to use Pythagoras, so split c into two so that we get two right-angled triangles.

By Pythagoras: $h^2 + x^2 = b^2$ and
$h^2 + (c-x)^2 = a^2$

Subtracting to eliminate h:

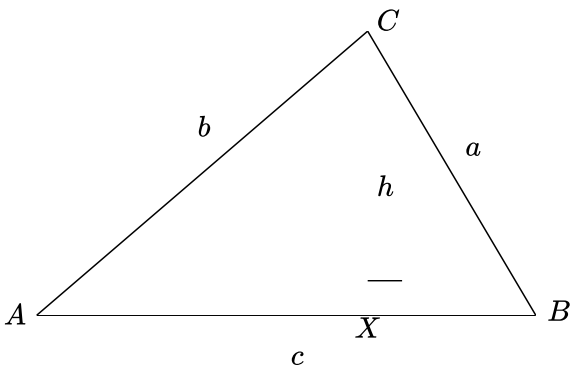
$x^2 - (c-x)^2 = b^2 - a^2$
$a^2 = b^2 + c^2 - 2cx$

But $x = b\cos A \therefore$

$a^2 = b^2 + c^2 - 2bc\cos A$

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 6 · Proof of Sine Rule



The idea is that we can use the common length of ΔACX and $\angle XBC$, i.e. h, to connect the two triangles, and therefore connect their angles/length.

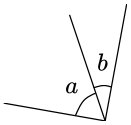
Using basic trigonometry:

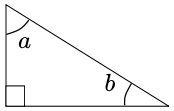
$h = a\sin B$ and $h = b\sin A$

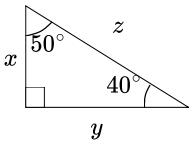
$\therefore a\sin B = b\sin A$

$\therefore \frac{a}{\sin A} = \frac{b}{\sin B}$

Chapter 9 · Original slide 7 · Just for your interest…







Why does “cosine” contains the word “sine”?

Complementary Angles

add to 90°

Therefore these angles are complementary.

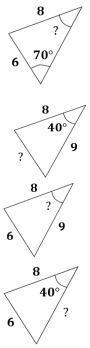
i.e. The cosine of an angle is the sine of the complementary angle.

Hence cosine = COMPLEMENTARY SINE

$\therefore \mathbf{cos\,(50) = sin\,(40)}$

1.39 Trigonometry: Introduction

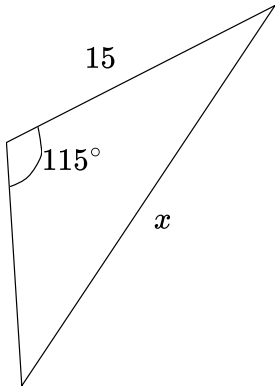
Chapter 9 · Original slide 8 · OVERVIEW: Finding missing sides and angles



When triangles are not right-angled, we can no longer use simple trigonometric ratios, and must use the cosine and sine rules.

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	
#2 Two sides known and a missing side opposite a known angle	Remaining side	
#3 All three sides	An angle	
#4 Two sides known and a missing side not opposite known angle	Remaining side	

Chapter 9 · Original slide 9 · Cosine Rule



We use the cosine rule whenever we have three sides (and an angle) involved.

b

Cosine Rule:

$a^2 = b^2 + c^2 - 2bc\cos A$

Proof at end of PowerPoint.

a A

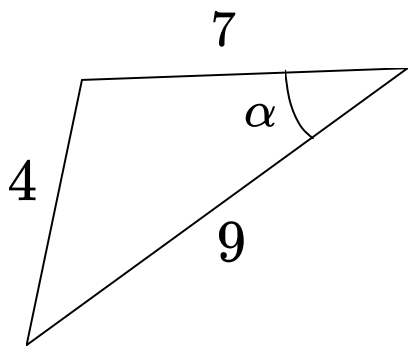
The only angle in formula is A, so label angle in diagram A, label opposite side a, and so on (b and c can go either way).

c

12

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 10 · Dealing with Missing Angles



You have	You want	Use
All three sides	An angle	Cosine rule

$\mathbf{a^2 = b^2 + c^2 - 2bc\cos A}$

Tip: The brackets are not needed, but students who forget about BIDMAS see $4^2 = \left(7^2 + 9^2 - 2 \times 7 \times 9\right)\cos\alpha$ and hence incorrectly simplify to $16 = 4\cos\alpha$

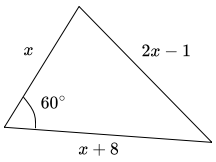
Label sides then substitute into formula.

Simplify each bit of formula.

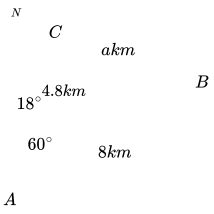
Rearrange "swapping" thing you're subtracting and result

Textbook Note: The textbook presents the rearrangement of the cosine rule: $\cos A = \frac{b^2+c^2-a^2}{2bc}$ to find missing angles. It is not advisable to memorise this

Chapter 9 · Original slide 11 · Harder Ones



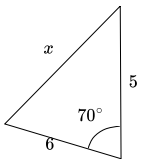
Determine the value of x.

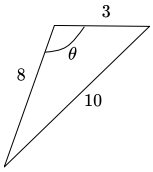


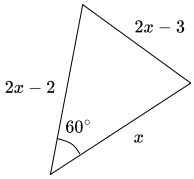
[From textbook] Coastguard station B is 8 km, on a bearing of $060°$, from coastguard station A. A ship C is 4.8 km on a bearing of $018°$, away from A. Calculate how far C is from B.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 12 · Test Your Understanding





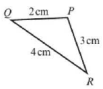


1 3 2

Note: You will get an obtuse angle whenever you inverse cos of a negative value.

Chapter 9 · Original slide 13 · Exercise 9A – Y1 pages 177-179

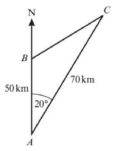
8 Show that $\cos P = -\frac{1}{4}$.



11 The lengths of the sides of a triangle are in the ratio $2 : 3 : 4$. Calculate the size of the largest angle.

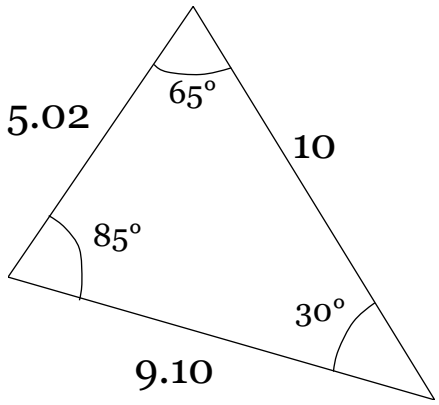
1.39 Trigonometry: Introduction

Chapter 9 · Original slide 14 · Exercise 9A – Y1 pages 177-179



17 The diagram shows three cargo ships, A, B and C, which are in the same horizontal plane. Ship B is 50 km due north of ship A and ship C is 70 km from ship A. The bearing of C from A is $020°$.
a Calculate the distance between ships B and C, in kilometres to 3 s.f. (3 marks)
b Calculate the bearing of ship C from ship B. (4 marks)

Chapter 9 · Original slide 16 · The Sine Rule



C B a

! Sine Rule:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

c b A

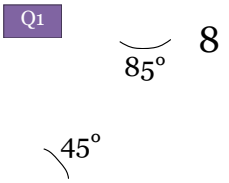
For this triangle, try calculating each side divided by the sin of its opposite angle. What do you notice in all three cases?

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

1.39 Trigonometry: Introduction

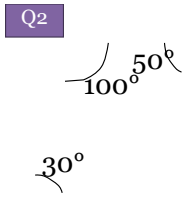
Chapter 9 · Original slide 17 · Examples

Q1



$$\frac{x}{\sin 85} = \frac{8}{\sin 45}$$
$$x = \frac{8 \sin 85}{\sin 45} = 11.27$$

Q2



$$\frac{x}{\sin 100} = \frac{8}{\sin 30}$$
$$x = \frac{8 \sin 100}{\sin 30} = 15.76$$

You have	You want	Use
#1: Two angle-side opposite pairs	Missing angle or side in one pair	Sine rule

Chapter 9 · Original slide 18 · Examples

Q3



$$\frac{\sin\theta}{8} = \frac{\sin 126^\circ}{10}$$
$$\sin\theta = \frac{8 \sin 126}{10}$$
$$\theta = \sin^{-1}\left(\frac{8 \sin 126}{10}\right)$$
$$= 40.33^\circ$$

Q4



When you have a missing angle, it's better to reciprocate to get:

$$\frac{\sin A}{a} = \frac{\sin B}{b}$$

i.e. in general put the missing value in the numerator.

$$\frac{\sin\theta}{5} = \frac{\sin 85}{6}$$
$$\sin\theta = \frac{5 \sin 85}{6}$$
$$\theta = \sin^{-1}\left(\frac{5 \sin 85}{6}\right)$$
$$= 56.11^\circ$$

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 19 · Exercise 9B – Y1 pages 181-183

8 Town B is 6 km, on a bearing of 020°, from town A. Town C is located on a bearing of 055° from town A and on a bearing of 120° from town B.
Work out the distance of town C from:
a town A $btownB$

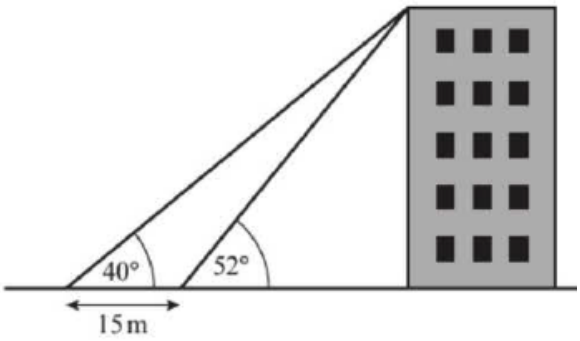
Chapter 9 · Original slide 20 · Exercise 9B – Y1 pages 181-183

11 In $\triangle ABC$, $AB = x$ cm, $BC = (4 - x)$ cm, $\angle BAC = y$ and $\angle BCA = 30^\circ$.
Given that $\sin y = \frac{1}{\sqrt{2}}$, show that

$x = 4(\sqrt{2} - 1)$

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 21 · Exercise 9B – Y1 pages 181-183

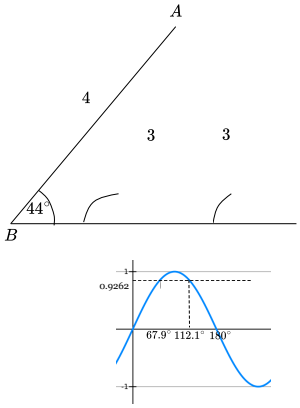


12 A surveyor wants to determine the height of a building. She measures the angle of elevation of the top of the building at two points 15 m apart on the ground.

a Use this information to determine the height of the building. (4 marks)

b State one assumption made by the surveyor in using this mathematical model. (1 mark)

Chapter 9 · Original slide 23 · The 'Ambiguous Case'



Suppose you are told that $AB = 4$, $AC = 3$ and $\angle ABC = 44°$. What are the possible values of $\angle ACB$?

Your calculator will give the acute angle of $67.9°$ (i.e. C_2). But if we look at a graph of sin, we can see there's actually a second value for $\sin^{-1}(0.9262)$, corresponding to angle C_1.

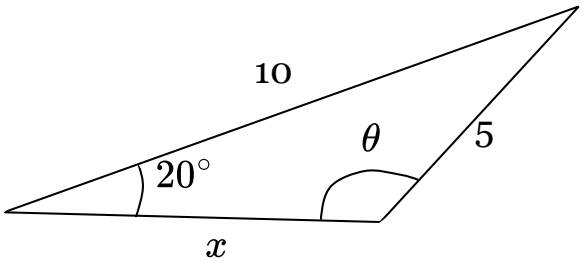
! The sine rule produces two possible solutions for a missing angle:

$\sin\theta = \sin(180° - \theta)$

Whether we use the acute or obtuse angle depends on context.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 24 · Test Your Understanding



Given that the angle θ is obtuse, determine θ and hence determine the length of x.

Chapter 9 · Original slide 25 · Exercise 9C – Y1 pages 184-185

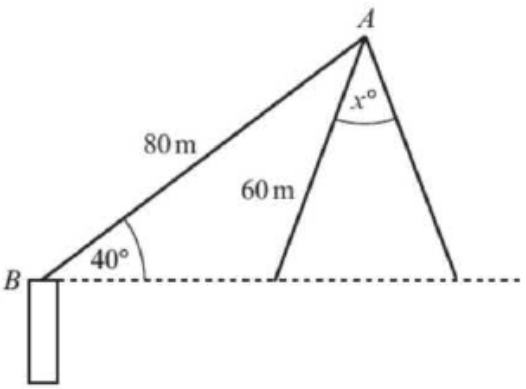
4 Triangle ABC is such that $AB = 4$ cm, $BC = 6$ cm and $\angle ACB = 36°$. Show that one of the possible values of $\angle ABC$ is $25.8°$ (to 3 s.f.). Using this value, calculate the length of AC.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 26 · Exercise 9C – Y1 pages 184-185

5 Two triangles ABC are such that $AB = 4.5$ cm, $BC = 6.8$ cm and $\angle ACB = 30°$. Work out the value of the largest angle in each of the triangles.

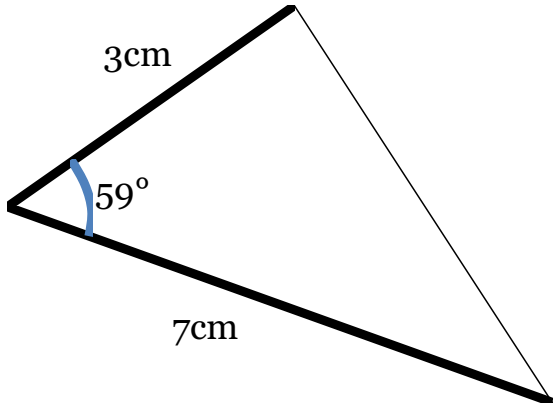
Chapter 9 · Original slide 27 · Exercise 9C – Y1 pages 184-185



6 a A crane arm AB of length 80 m is anchored at point B at an angle of $40°$ to the horizontal. A wrecking ball is suspended on a cable of length 60 m from A. Find the angle x through which the wrecking ball rotates as it passes the two points level with the base of the crane arm at B. (6 marks)
b Write down one modelling assumption you have made. (1 mark)

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 28 · Area of Non Right-Angled Triangles

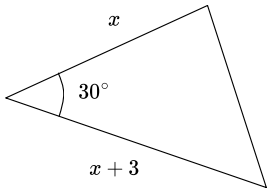


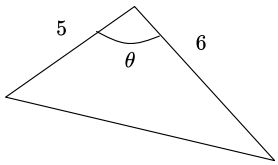
Area = $\frac{1}{2}ab\sin(C)$

where C is the angle between two sides a and b.

Tip: You shouldn't have to label sides/angles before using the formula. Just remember that the angle is between the two sides.

Chapter 9 · Original slide 29 · Test Your Understanding





The area of this triangle is 10.

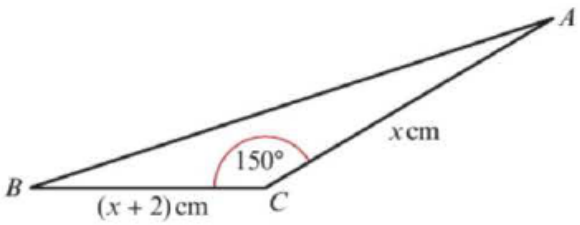
Determine x.

The area of this triangle is also 10.

If θ is obtuse, determine θ.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 30 · Exercise 9D – Y1 pages 186-187



4 In triangle ABC, $BC = (x + 2)$ cm, $AC = x$ cm and $\angle BCA = 150°$. Given that the area of the triangle is 5 cm^2, work out the value of x, giving your answer to 3 significant figures.

Chapter 9 · Original slide 31 · Exercise 9D – Y1 pages 186-187

5 In $\triangle PQR$, $PQ = (x + 2)$ cm, $PR = (5 - x)$ cm and $\angle QPR = 30°$. The area of the triangle is A cm^2.

a Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

b Use the method of completing the square, or otherwise, to find the maximum value of A, and give the corresponding value of x.

1.39 Trigonometry: Introduction

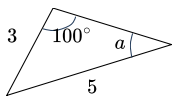
Chapter 9 · Original slide 32 · Exercise 9D – Y1 pages 186-187

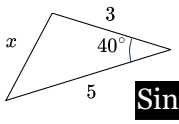
6 In $\triangle ABC$, $AB = x$ cm, $AC = (5 + x)$ cm and $\angle BAC = 150°$. Given that the area of the triangle is $3\frac{3}{4}$ cm^2

a Show that x satisfies the equation $x^2 + 5x - 15 = 0$.

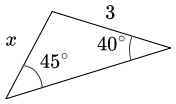
b Calculate the value of x, giving your answer to 3 significant figures.

Chapter 9 · Original slide 33 · Sin or cosine rule?

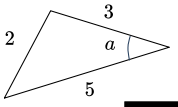




Sine



Sine Cosine



Sine

Recall that whenever we have two “side-angle pairs” involved, use sine rule. If there’s 3 sides involved, we can use cosine rule. Sine rule is generally easier to use than cosine rule.

Sine

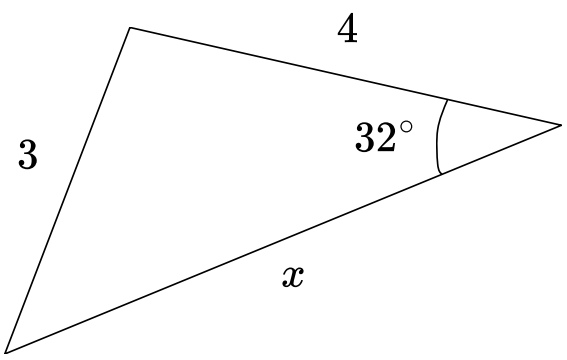
Cosine

Cosine

Cosine

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 34 · Using sine rule twice



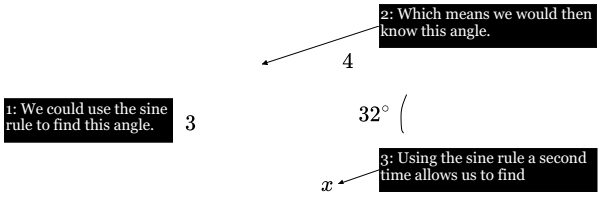
You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

Given there is just one angle involved, you might attempt to use the cosine rule:

This is a quadratic equation!

It's possible to solve this using the quadratic formula (using $\mathbf{a=1, b=-8\cos 32, c=7}$). However, this is a bit fiddly and not the primary method expected in the exam...

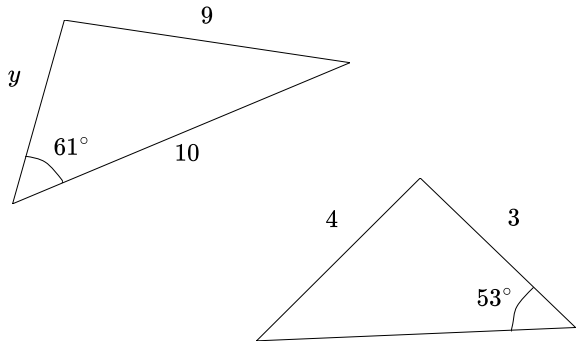
Chapter 9 · Original slide 35 · Using sine rule twice



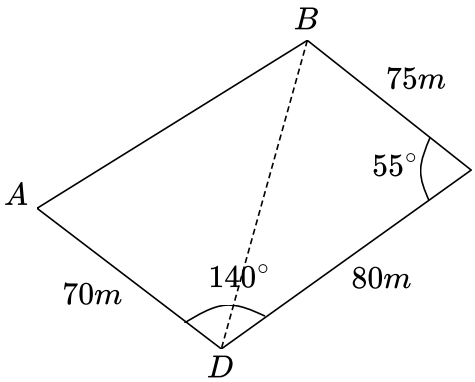
You have	You want	Use
#4 Two sides known and a missing side not opposite known angle	Remaining side	Sine rule twice

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 36 · Test Your Understanding



Chapter 9 · Original slide 37 · Problem Solving With Sine/Cosine Rule



[From Textbook] The diagram shows the locations of four mobile phone masts in a field, $BC = 75m$.

$CD = 80m$, angle $BCD = 55°$ and angle $ADC = 140°$.

In order that the masts do not interfere with each other, they must be at least 70m apart.

Given that A is the minimum distance from D, find:

a) The distance A is from B

b) The angle BAD

c) The area enclosed by the four masts.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 38 · Exercise 9E – Y1 page 189-191

12 The perimeter of $\triangle ABC = 15$ cm. Given that $AB = 7$ cm and $\angle BAC = 60^\circ$, find the lengths of AC and BC and the area of the triangle.

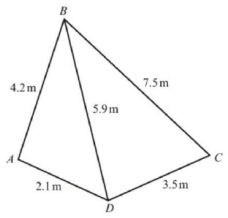
Chapter 9 · Original slide 39 · Exercise 9E – Y1 page 189-191

13 In the triangle ABC, $AB = 14$ cm, $BC = 12$ cm and $CA = 15$ cm.
a Find the size of angle C, giving your answer to 3 s.f.
b Find the area of triangle ABC, giving your answer in cm^2 to 3 s.f.

1.39 Trigonometry: Introduction

Chapter 9 · Original slide 40 · Exercise 9E – Y1 page 189-191

14 A flower bed is in the shape of a quadrilateral as shown in the diagram.
a Find the sizes of angles DAB and BCD. (4 ma...
b Find the total area of the flowe
(3 ma...
c Find the length of the diagonal



Chapter 9 · Original slide 41 · Exercise 9E – Y1 page 189-191

15 $ABCD$ is a square. Angle CED is obtuse.
Find the area of the shaded triangle. (7 marks)

